

Donghwan Kim

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EDUCATION

The Pennsylvania State University

State College, PA Aug 2024 – Present

Doctor of Philosophy, Computer Science and Engineering

Advisor: Kiwan Maeng

Seoul National University (SNU)

Seoul, Korea Mar 2022 – Feb 2024

Master of Science, Interdisciplinary Program in Artificial Intelligence

Advisor: Jung Ho Ahn

Seoul National University (SNU)

Seoul, Korea Mar 2016 – Feb 2022

Bachelor of Science, Electrical and Computer Engineering, *cum laude*

Two-year leave for military service

RESEARCH INTERESTS

My research focuses on building efficient computer systems through hardware–software co-design. My work spans privacy-preserving machine learning, including differentially private training and fully homomorphic encryption, and recently extends to LLM serving systems for agentic AI workloads.

PUBLICATIONS

- [IISWC'26] **Donghwan Kim**, Prakhar Singh, Younghoon Min, Jongryool Kim, Jongse Park, Kiwan Maeng, “Characterizing How Complex Agentic AI Systems Handle General Tasks: A Trace-Based Simulation Study,” *IEEE International Symposium on Workload Characterization*, 2026 (*Best Paper & Distinguished Artifact Awards*)
- [OSDI'26] **Donghwan Kim**, Xin Gu, Jinho Baek, Timothy Lo, Younghoon Min, Kwangsik Shin, Jongryool Kim, Jongse Park, Kiwan Maeng, “Cocoon: A System Architecture for Differentially Private Training with Correlated Noises,” *USENIX Symposium on Operating Systems Design and Implementation*, 2026
- [CAL'25] SeokHyeon Kong, **Donghwan Kim**, Euseong Seo, Kiwan Maeng, “Characterizing the System Overhead of Discrete Noise Generation for Differential Privacy,” *IEEE Computer Architecture Letters*, 2025
- [CCS'24] Jae Hyung Ju^{*}, Jaiyoung Park^{*}, Jongmin Kim, Minsik Kang, **Donghwan Kim**, Jung Hee Cheon, Jung Ho Ahn, “NeuJeans: Private Neural Network Inference with Joint Optimization of Convolution and FHE Bootstrapping,” *ACM SIGSAC Conference on Computer and Communications Security*, 2024 (**equal contribution*)
- [Access'23] **Donghwan Kim**^{*}, Jaiyoung Park^{*}, Jongmin Kim, Sangpyo Kim, Jung Ho Ahn, “HyPHEN: A Hybrid Packing Method and Its Optimizations for Homomorphic Encryption-based Neural Networks,” *IEEE Access*, 2023 (**equal contribution*)
- [DISCC'23] Jaiyoung Park, **Donghwan Kim**, Wonkyung Jung, Sangpyo Kim, Jongmin Kim, Jung Hee Cheon, Jung Ho Ahn, “Toward Practical Privacy-Preserving Convolutional Neural Networks Exploiting Fully Homomorphic Encryption,” *Workshop on Data Integrity and Secure Cloud Computing*, 2023
- [ISCA'23] Jongmin Kim, Sangpyo Kim, Jaewan Choi, Jaiyoung Park, **Donghwan Kim**, Jung Ho Ahn, “SHARP: A Short-Word Hierarchical Accelerator for Robust and Practical Fully Homomorphic Encryption,” *International Symposium on Computer Architecture*, 2023

AWARDS & HONORS

Best Paper Award & Distinguished Artifact Award

IISWC 2026

WORK EXPERIENCE

LLM Serving System Research Intern

San Jose, CA May 2026 – Aug 2026

SK Hynix America

- Built an LLM serving simulator for agentic AI workloads
- Explored CXL 3.0 memory pooling for agentic workloads

RESEARCH EXPERIENCE

Graduate Research Assistant

State College, PA Aug 2024 – Present

The Pennsylvania State University (Advisor: Kiwan Maeng)

- *Distributed training systems*: developing a scalable DP-SGD training framework on DeepSpeed and PyTorch to optimize multi-GPU gradient aggregation

Graduate Research Assistant

Seoul, Korea Mar 2022 – Feb 2024

Seoul National University (Advisor: Jung Ho Ahn)

- *Algorithm optimization*: devised convolution algorithms and data packing formats that minimize expensive homomorphic encryption operations
- *AI accelerator design*: designed a domain-specific architecture for homomorphic encryption operations

TUTORIALS

“CrypTorch: Easy-to-use and Efficient PyTorch-based Compiler Infrastructure for Machine Learning with MPC,”
ASPLOS 2026 Tutorial [link]

TEACHING EXPERIENCE

Introduction to Computer Architecture (CMPEN 431)

Fall 2025

Teaching assistant — weekly recitations, assignment grading

Computer Organization and Design (CMPEN 331)

Fall 2024

Teaching assistant — weekly recitations and office hours, assignment and exam grading

Programming Methodology

Spring 2022

Teaching assistant — weekly lab sessions, C++ final term project

SCHOLARSHIPS

AI Fellowship Scholarship

2022

Seoul National University, Dept. of Engineering

Merit-based Scholarship

2016, 2017, 2020, 2021

Seoul National University, Dept. of Engineering

Hansung Sonjaehan Scholarship

2015

Hansung Sonjaehan Foundation — \$4,000 for gifted students in science and engineering

TECHNICAL SKILLS

Machine learning systems: PyTorch, DeepSpeed, vLLM, Dynamo

Programming languages: Python, C/C++, CUDA, Verilog

Algorithms & math: Differentially private SGD (DP-SGD), Fully Homomorphic Encryption (CKKS)